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The sandiness of sand

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ABSTRACT: This paper addresses the exploration of the local sandstone Marés, used by Jørn Utzon (JU) in his construction of Can Lis, Mallorca 1972. This study not only investigates scientific properties but also how it functions as a material deeply embedded in a complex matrix of architectural ethics, aesthetics, and culture. The paper also delves into the way JU's vision of the material was not purely functional but embraced its inherent properties of decay, allowing the building to return to nature over time. Through a residency 2023, talks with the local architects involved in the restauration and direct observations living close to nature for several weeks in Can Lis, it has been documented that the special and unique atmosphere is strongly related to the porosity and weaknesses of the sandstone. Evidence in the detailing of the house by JU supports the idea that he was aware of this contradiction.

1 SEEING MATERIALS FOR WHAT THEY WERE

"We were concerned with the seeing of materials for what they were: the woodness of wood; the sandiness of sand. With this came a distaste of the simulated, such as the new plastics of the period – printed, coloured to imitate a previous product in "natural" materials. Dislike for certain mixes, particularly with technology, such as the walnut dashboard in a car. We were interested in how things could be with technology touching every- thing and everyone. We foresaw a general reappraisal of values would occur, since as we "read" through the aspiration-images on offer in the magazines, the approach of the acquisitive society" (Smithson 1990, p. 201).

This paper draws on the ideas of Alison and Peter Smithson, particularly their reflection on the "sandiness of sand" which they coined to challenge the simulated aesthetics of modern consumerist society. The Smithsons' notion of valuing materials for what they are, without the superficiality of synthetic replication, resonates strongly with JU's approach in Can Lis. JU, like the Smithsons, exhibited in Mallorca a distaste for modern materials that mimicked natural products without capturing their essence. By focusing on the authentic properties of marés, JU positioned himself against the increasingly prevalent trends of post-war consumerism that favoured cheap, artificial materials. In Can Lis, the natural weathering and decay of the marés sandstone was not something to be combated but embraced as an inevitable and beautiful part of the building's life cycle. In his paper for ICSA 2022: "An ecopoetic architect" Nicolai Bo Andersen also unfolds different sources to JU´s critical use of materials: "Similarly, in the relation of material to technique, the architect does not impose form on matter, but rather allows the materials to show what they can do." (Andersen 2022, p. 334-335).

This philosophical stance also ties into contemporary critiques of modernity and its role in alienating humans from the natural world, a theme that resonates throughout the paper. As told in the Smithson's quote –written at the end of the eighties and published in 1990, Can Lis, designed and built between 1970-1974– the Smithsons looked back at their work from the early 1950's. They say they foresaw a change in values in post-war society and were concerned with seeing materials for what they were as opposed to the material simulacrum of cheap post-war materials. JU's decision to use Marés sandstone in Can Lis is explored not only from a practical

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perspective but as a deliberate aesthetic and philosophical choice. Marés, a porous and fragile stone, is highly susceptible to the effects of weathering. However, JU's bold vision for Can Lis included allowing this process of natural decay to occur, integrating the building into the landscape in a profound way. In this light, JU's use of Marés was more than a pragmatic decision based on the availability of materials. It reflected his deep understanding of architecture as a living entity, one that is subject to the forces of nature and time. In this way, Can Lis designed not just to coexist with its environment but to eventually dissolve back into it, making the process of decay an integral part of the building's aesthetic and functional lifespan. This approach to materiality challenges conventional notions of permanence in architecture, highlighting the impermanence of all things, including buildings. We view this as a lens for looking at architecture today when modernity is being arrested for complicity in the now far advanced ecological crisis.

But what did The Smithsons mean by saying "a distaste of the simulated" and which "reappraisal of values" did they have in mind? A reflection that echoes todays imperative rethinking of our building practise. A crucial philosophical critique also expressed by the political theorist and philosopher Jane Bennett: "It hit me then in a visceral way how American materialism, which requires buying ever-increasing numbers of products purchased in ever shorter cycles, is antimateriality. The sheer volume of commodities, and the hype consumptive necessity of junking them to make room for new ones, conceals the vitality of matter" (Bennett 2010, p. 5). In his book "The uncontrollability of the world" Hartmut Rosa diagnose late modern society in its present state of ecological crisis, democratic crisis and mental stress crisis: "Modernity stands at risk of no longer hearing the world, and for this very reason, losing its sense of itself. This is the conclusion of my sociology of our relationship to the world. Modernity has lost its ability to be called, to be reached" (Rosa, 2018, p. 31). His term uncontrollability has more than one meaning. On one side he describes how we in the modern world attempt to take full control of the world, possessing and exploiting it to a degree that the world has returned in the form of an uncontrollable monster. On the other hand, he shows us that it is only by embracing that which is uncontrollable, unpredictable and unforeseen that we feel truly alive. Rosa calls for alternatives and for us to relate to the world and achieve a state of resonance where we again can feel touched, moved and alive. "The basic mode of vibrant human existence consists not in exerting, control over things but in resonating with them, making them respond to us" (Rosa 2018, p. 31).

Departing from Rosa´s analysis, we speculate: What are the processes of making and the material qualities that connects us to matter and materials in a way that we can (again) feel touched, moved and alive? We study Can Lis and the use of the local Sandstone by Utzon including site observations to exemplify the importance of Bennetts material vitality and Rosa ´s call for resonance. Aesthetic values that could be associated with a resistance to this alienation of the late modern society.

2 GEOLOGY AND ORIGIN

"Those who had resources, they built the houses with better marés and left it visible. If they had no resources, they used a worse quality marés and covered it with lime mortar……Can Lis was built by Miquel Vidal, a building contractor who made village houses and animal stables in the countryside. The execution of Can Lis is more like a sheep stable than a village house" (Interview with Francisco Cifuentes in Ruiz Allan 2022) & "Holiday homes, when out of season in the winter, are usually very ugly. They seem a bit abandoned, decadent. Coastal sites have a deplorable appearance in winter. Yet, you visit this unoccupied house (Can Lis) in the winter, and it is wonderful. It is a homage even in the absence of its owners. The table, the chairs, the columns, the walls, the pavements, etc., they are imprints that can withstand the solitude of the house" (Interview with Luis Santa-María in Ruiz Allan 2022).

The marés sandstone is a stone of the Balearic Islands which is extracted in blocks of 40x40x80cm, and it is very useful for the execution of walls and vaults. Their characteristics depend on their origin which can be calcarenite (cemented sand grains with calcium from

mollusks shells) or eolianite (sedimented sand) and therefore each quarry has very different properties with resistances which vary between 25 and 630Kg/cm².

Newer scientific research documents origin, history, properties, quarries and qualities currently available of the local sandstone marés (Sánchez Cuenca 2010). Illustrated below is the maps of prehistoric Mallorca (Map by Guillem Colom from Sánchez Cuenca 2010) and another one over layering a geological map with that of the principal location of quarries (Mas Gornals 2017, p. 415). The overall idea of máres and its origin is conceptually understood by reading Colom´s map because millions of years ago inland Mallorca in between the mountains Serra de Tramuntana northwest and Serra de Llevant southeast as well as the today coastal areas was all covert by the Mediterranean Sea. Principally the marés come from these areas between the mountains and the areas bordering the mountains. This again explains the etymology of marés meaning "belonging to or coming from the sea".

The complexity increases when different kinds of stone all named marés can have very different composition and mixture of marine sedimentation and continental sedimentation as well as having very different age ranging from less than 1 million years to more than 10 million years. (Mas Gornals 2017, p. 414). To illustrate the problems of classification Guillem Mas Gornals points out that the so-called Santanyí Stone is included in his study of marés, since, despite being traditionally considered a rock different from marès, it generally shares common characteristics with it, since it is also a calcarenite, easy to work with and traditionally used in construction (Mas Gornals 2017, p. 415). According to interviews on site (Can Lis 2023) with Francisco Cifuentes the J.U plateau; the floor of Can Lis is constructed from the Santanyí stone (Piedra de Santanyí) while the walls are from a younger and more porous type of marés. Prior to modern research and at the time when JU readdressed the use of marés in his contemporary construction of Can Lis the selection and appliance of marés for construction was to a large degree dependent of tacit knowledge in the quarries and among experienced builders. J. U would also have been able to visit historic sandstone buildings In Mallorca ranging from the most ancient Castell de Bellver and La Llotja, the old stock exchange building, both from the 15th century to more recent village house and sheep stables.

In his phd. dissertation "La Lonja de Guillem Sagrera. El salón de los Mercaderes" Francisco Cifuentes revels the history of local construction that is necessary to fully understand the use of different stone qualities: "The Santanyí Stone is a sandstone but as it has isolated pores it has a much greater hardness than marés, which has open pores. Traditionally, Santanyí Stone was used in sculpture given the hardness and durability of the material. However, marés was usually used for construction since the Santanyí Stone had a large extra cost both for cutting in the quarry and for transporting the material (to Palma red.). This stone is found on the coast of Felanitx and Santanyí. . .." (Cifuentes 2015, p. 257).

Now anyone who is interested in the architectural qualities of building with local sandstone in Mallorca can visit, see and touch the facades of both La Lonja completed 1457 as well as Can Lis completed 1971. Both places you will that the facades have a beautiful weathering or patina as in very uneven surfaces, eroding or experiencing decay. So boldly you could say that the "Piedra de Santanyí" will behave similarly to the marés of Can Lis after standing outside for more than five hundred years.

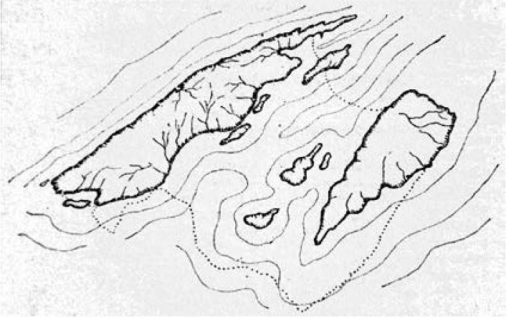


Figure 1. Map by Guillem Colom from Sánchez Cuenca 2010.

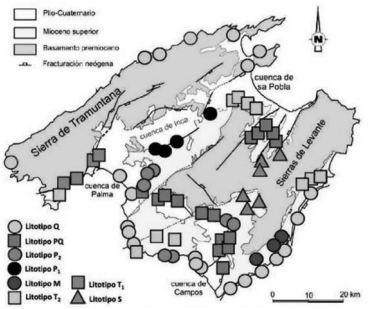


Figure 2. Layering a geological map with that of the principal location of quarries. Ill. MAS GORNALS 2017.

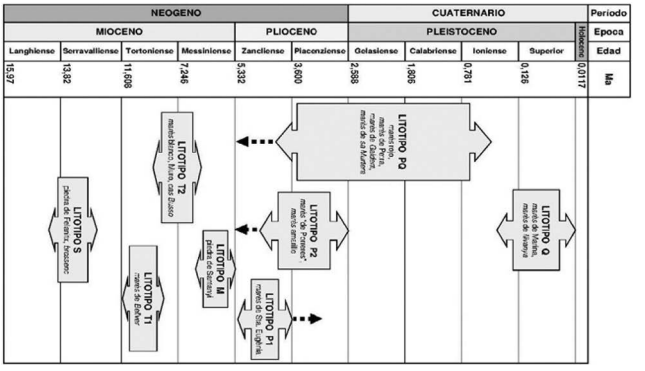


Figure 3. Chronostratigraphi. Marés from Mallorca. Ill. by Mas Gornals 2017.

3 THE WALL, THE WINDOW AND THE BENCH

3.1 *First-hand experiences avoiding the trap of prevailing mythologies.*

Being educated as a Danish architect, or a Spanish architect, you properly will study JU including Can Lis. But how do we distinguish literature and observation being made from a distance, those based on shorter site visits and those based on dwelling there and taking time to experience the architecture with all your senses day and night. How can we understand and discuss important issues connected to the material, the atmosphere and what Rosa names resonance without experiencing it first-hand? At least our residency 14 days in April 2023 made us think of that and the difference we felt to earlier short visits. And the following observations and the reflections: the wall, the window and the bench were what came out of that.

Recent research has questioned prevailing mythologies of Can Lis and what we may have learnt from leading historians and publications: "Numerous authors, including Weston, Faber, Pardey and others have written of Utzon's involvement on site without properly qualifying the relation of this involvement to the significant documentation of the house to which he always returned and thus has given a misleading impression that Utzon's virtually built the house himself from sketches. Our interviews revealed that great emphasis was placed by Utzon on fully solving his projects by drawing and redrawing". (Reinmuth 2015, p. 42). Avoiding the trap of prevailing mythologies is easier said than done. Many different stories have been told, greater part of the Utzon archives is not available, or cataloguing are still pending. But the following observations is based on our own first-hand experiences and an interview with the local architect involved in the restauration.

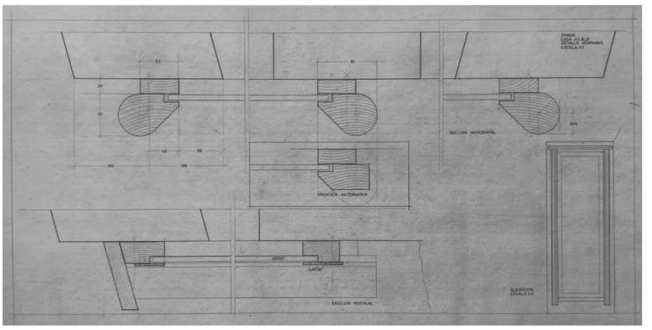


Figure 4. Detail window frame. Can Lis. © Utzon Arkivet.

3.2 *The wall and the window*

Can Lis, April 6, 2023, 12:06 pm: "A fly caught in a spider's web allows the soft stone to weather, falling in front of the view towards the sea." (Extract from the art project by Hartmann & Drewniak 2023). After engaging in dialogue with a local mason and builder, JU decided to frame the view towards the sea through a series of 11 windows, each shaped as rectangular cone carved from the marés stone. From the interior, these windows appear as deep cutouts in a two-meter-thick wall. However, when viewed externally, one can see that these spaces are small, with thin walls that extend beyond the room itself, protected by an all-encompassing overhang. This overhang functions much like a sun visor, shielding the windowpanes from excessive glare. JU drew inspiration from his visit to the sandstone cave beneath the cliff where the house stands. His translation of this experience into the architecture is evident in his deliberate exaggeration of the wall's thickness, as well as in the positioning of the window frame on the exterior, rendering it invisible from within. This unique detail is what grants Can Lis its distinctive and often debated spatial significance within the broader context of architectural history. Yet, despite the extensive discussion this feature has received, might there be more to uncover through closer, meditative observations? In April 2023, we embarked on such observations, leading us to consider whether there is a deeper layer to this story, one previously overlooked. Our focus centres on the relationship between the wall and the window—not just in terms of their physical joining, but in how these two elements are separated, and what emerges in the space between them. Notably, JU did not merely place the window frame on the exterior of the structure. Instead, he offset it approximately sixty millimetres from the edge of the sandstone, creating a deliberate gap between the window and the wall. This gap, we argue, serves as a space where the passage of time and the inevitability of decay are not only acknowledged but embraced. JU's design anticipates this future weathering, incorporating it into the architecture itself—even when the progression of decay may have outpaced his original expectations. When cutting marés stone, one can achieve a relatively straight edge, although it is never perfectly linear due to the inherent softness and tolerance of the material. This characteristic becomes a critical consideration in JU′s design. By April 2023, the process of decay and weathering had become more pronounced, particularly at the corners of the stonework. The edge of the sandstone, which delineates the space inside, now draws a visible line of decay. In this way, JU's encounter with the cave below the house has become even more evident. This element, less a cultural interpretation and more a natural reflection of time's effects, underscores the inherent "perfect imperfection" of natural stone. In some areas the decay has advanced to such a degree that a hand can be placed between the stone and the window frame. This reveals an important insight: the window frame could not have been installed on the interior without compromising the space for this natural decay process. JU's solution balances spatial excellence with practicality, reflecting an aesthetic philosophy that does not distinguish between the organic progression of decay and the materiality of the structure itself.



Figure 5. Photos window frame. Can Lis 2013. © Frans Drewniak.

3.3 *The wall and the bench*

The second observation centers on how JU integrates the wall and the bench within the courtyard. Can Lis, April 12, 2023, 13:03 pm: "Behind the tiled bench, a narrow channel separates the furniture from the wall. We sweep the gutter with the small, round Mallorca broom we had just purchased from the old grocery store in Cas Concos." (Extract from the art project by Hartmann & Drewniak 2023). In this instance, the separation between the wall and the bench is particularly pronounced. The backrest of the bench is supported by cantilevered marés blocks extending from the wall, while the seat rests on blocks that elevate it above the floor structure. Our close engagement with the physical elements and their materials was catalysed by the art project we were developing on-site. Having just acquired a traditional, small, round Mallorca broom, we began to draw in the sand within the courtyard, which led us to investigate the gap between the wall and the bench. However, what captured our interest was not merely the presence of the gap itself, but what lay within it. We soon realized that the gap functioned as a gutter, collecting the sand falling from the eroding sandstone wall. When we swept the gutter, the broom fit perfectly, as though designed for the task. The result was two full handfuls of sand—a physical manifestation of the ongoing decay. The separation between the bench and the wall is not a novel observation. In the sandstone dormitories at Can Lis, for instance, the beds were painted white to prevent dust from accumulating on the linens. What we propose in this context is that the separation here reflects Jorn Utzon's meticulous consideration of both functional and aesthetic concerns. The gap serves not only as a practical feature, accommodating the natural processes of erosion and decay, but also as a forward-thinking design solution that anticipates these changes. During a walk with Francisco Cifuentes in April 2023 at Can Lis, he pointed out the exterior wall between two bedrooms. According to Cifuentes, this wall was added by Jørn Utzon before moving to Can Feliz in 1994. Upon closer inspection, it becomes apparent that this external wall was constructed using a harder type of sandstone, likely Santanyí stone, in contrast to the material used in the rest of the walls. For the first time, we began to realize that had the house been built entirely from this material, its overall aesthetic and atmosphere would have been more monumental. Instead, the use of softer materials imparts a simplicity to the space, which Cifuentes aptly describes as resembling a "sheep stable" in the interview "Patterns of Life" (Ruiz Allan 2022).

4 CONCLUSION

We propose that the efforts to repair and the visible traces of use and wear over time can be seen as a response to the alienation inherent in late-modern consumer culture. These efforts

reflect attempts to achieve resonance, as articulated by Hartmut Rosa. Buildings and building materials should be understood as integral parts of ongoing natural processes, as eloquently described by Mostafavi and Leatherbarrow in their book on weathering: "Paradoxically, weathering produces something already there by subtraction. This redefines the relationship between art and nature. In design, art is traditionally seen as the force that shapes nature; yet, over the lifespan of a construction, nature reclaims and reshapes the 'finished' artwork… This is the ultimate assimilation of an artwork back into its original location, the place from which it was first drawn." (Mostafavi & Leatherbarrow 2001, p. 64-65). This concept helps explain the bold claim made by Rafael Moneo during his 2018 speech at the Utzon Center in Aalborg. Moneo argued that JU′s choice of marés, a soft, locally sourced sandstone, was a deliberate decision. By selecting a material that would naturally erode over time, Utzon embraced the idea that his building would eventually dissolve and return to the seashore cliff from which it originated. The advanced state of decay observed at Can Lis when local architects Aulets documented the building prior to its renovation in 2011-2012 was likely far more extensive than Utzon himself had anticipated. However, JU's detailed design choices reveal a deep awareness of the porosity and inevitable deterioration of the soft sandstone.

JU was deeply influenced by his time working with Alvar Aalto in the 1930s. Like Aalto's summer home in Muuratsalo, Finland—an experimental house where Aalto explored new ideas with bricks and other materials—Can Lis stands as a radical experiment. Positioned at the edge of a cliff and exposed to the harsh sea climate, the house offers an intimate yet serene experience, much like a temple. The beauty of the dwelling is inseparable from the fragility of the same sandstone on which it rests. As Caitlin DeSilvey notes: "Strategies to arrest decay always destroy some cultural traces, even as they preserve others. And decay itself may clear a path for certain kinds of remembrance despite (or because of?) its destructive energies." (DeSilvey 2017, p. 33). In the case of Can Lis, this "certain kind of remembrance" may be tied to the concept of provenance:



Figure 6. Photo mares wall Can Lis 2013. © Frans Drewniak.

where do materials come from, and what histories do they carry? The local marés sandstone, the main protagonist in this story, is geologically young, having formed just four million years ago (Plioceno. See Figure 3) from an ancient seabed. Exposed to the mist of the same sea, the stone erodes and returns to sand at a rate far faster than one might expect—much more rapidly than buildings constructed inland.

In this analysis, we have focused on two key architectural elements: the relationship between the wall and the window, and that between the wall and the bench. Our aim has been to demonstrate that JU designed Can Lis with decay in mind, embedding it into the construction and the joints. The pace of this decay, however, may have exceeded even his expectations. As Jan Utzon reflected in an interview: "And traditionally no one has ever lived along the coast anyway… So, it's a harsh climate down here as opposed to further inland where nothing happens. It's like parking your car by the sea or parking it inland." (Espanyol et al. 2025). Today, many might argue that building directly on the seashore should be avoided to protect fragile natural landscapes. According to interviews with JU´s family, after several years at Can Lis, JU and Lis Utzon found the winters along the coast too humid and eventually moved to the drier, more comfortable Can Feliz, where the term "comfort" was used in its traditional sense as a form of shelter from the elements. It could even be suggested that JU, by building Can Lis as part of the seashore cliff, sought to expel himself from the comforts of modern living, rather than protect and shelter the building from its natural surroundings. Such an interpretation could help us address a seemingly impossible question: How can we preserve this cultural heritage when the ongoing decay is one of the most significant aspects to be preserved?

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